ELECTRICITY AND MAGNETISM; ELECTROMAGNETIC WAVES

Almost every technology used in daily life—telephones, the internet, electrical systems, and even light itself—is based on a single fundamental physical framework: **electromagnetism**.

Electricity and magnetism were long considered separate phenomena. However, studies conducted in the 19th century revealed that these two domains are, in fact, different manifestations of the same underlying physical reality. This unification forms one of the most powerful theories in modern physics.

Electricity

Electricity forms the foundation of most modern technological systems. The fundamental constituents of matter are fermionic charges, primarily protons, neutrons, and electrons. At the core of electrical phenomena lies the interaction between these charged particles.

Neutrons are electrically neutral, while protons carry a positive charge and electrons carry a negative charge. Opposite charges attract each other, whereas like charges repel. The electrostatic attraction between protons and electrons is what holds atoms together and ultimately gives rise to matter.

Coulomb's Law

Coulomb's Law describes the inverse-square interaction between point charges in vacuum. According to this law, the force between two point charges is directly proportional to the product of their charges and inversely proportional to the square of the distance between them.

In mathematical form:

$$F = k \cdot \frac{q_1 \cdot q_2}{r^2}$$

This inverse-square dependence arises from the spherical symmetry of three-dimensional space. The law applies strictly to static charges and neglects relativistic effects.

Coulomb's Law can be interpreted as a special solution of Gauss's Law for systems with spherical symmetry:

$$\nabla \cdot \mathrm{E} = \frac{\rho}{\varepsilon_0}$$

This equation demonstrates that electric fields originate from charge distributions.

Electric Field

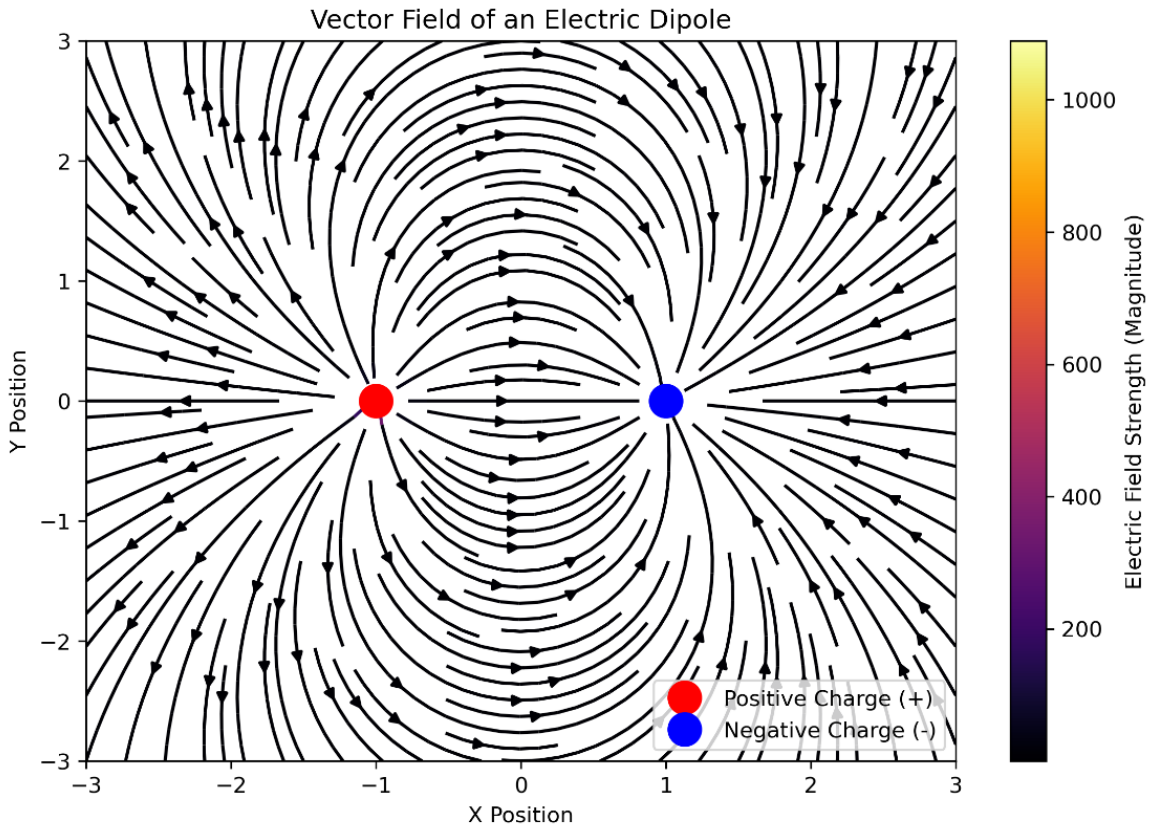
The region of influence created by electric charges is called the **electric field**. It is a vector quantity denoted by **E** and describes both the magnitude and direction of electric interactions.

The electric field is defined as the force per unit positive test charge:

$$E = \frac{F}{q}$$

Field lines originate from positive charges and terminate at negative charges. The density of these lines represents the strength of the field.

To further illustrate this concept, the behavior of electric field lines can be simulated using an electric dipole model. In the Python vector field simulation developed for this project, based on Coulomb's Law and the principle of superposition, it is clearly observed that the field strength increases significantly near the charges (highlighted regions).



Electric Potential and Electric Current

A charge in an electric field possesses potential energy depending on its position. This quantity is expressed through **electric potential**, defined as energy per unit charge and measured in volts.

The potential difference between two points is called **voltage**. It represents the energy gradient that drives charge motion from higher potential to lower potential.

In conductive materials, free electrons move under the influence of this potential difference, producing a **current**, defined as the directed flow of charge. Voltage is the driving force behind this motion.

A battery establishes a potential difference that initiates and sustains current in a circuit. As voltage increases, the electric field strength increases, resulting in stronger electron acceleration and higher current.

Resistance in Electrical Circuits

An electrical circuit consists of conductive paths and elements that regulate current flow. The opposing factor to current is **resistance**, which describes how strongly a material resists electron flow.

Electrons do not move in straight trajectories through a conductor; instead, they continuously collide with atomic structures, which limits current.

Resistance depends on the length, cross-sectional area, and material properties of the conductor. Current is directly proportional to voltage and inversely proportional to resistance.

Without resistance, circuits would be uncontrollable and could lead to damage of components.

Ohm's Law

Ohm's Law, formulated by Georg Simon Ohm in the 19th century, describes the relationship between voltage, current, and resistance:

$$V = I \cdot R$$

Resistance is measured in ohms (Ω). A higher resistance corresponds to lower current for a given voltage.

This linear relationship allows any one of the three variables to be determined if the other two are known.

On a microscopic scale, Ohm's Law can be expressed as:

$$\mathrm{J} = \sigma \mathrm{E}$$

This formulation represents the macroscopic result of electron scattering processes in conductors.

Capacitors

Capacitors are circuit elements that store electrical energy in the form of an electric field. Their unit is the farad (F). A capacitor consists of two conductive plates separated by an insulating material (dielectric).

The dielectric prevents direct charge flow while allowing energy storage in the electric field between the plates.

When voltage is applied, equal and opposite charges accumulate on the plates, creating an electric field. No direct current flows through the dielectric.

Capacitors are widely used in filtering, resonance circuits, and power regulation systems due to their ability to store and release energy efficiently.

Magnetism and Magnetic Fields

Electricity and magnetism are deeply interconnected aspects of physics. Moving electric charges generate electric currents, and these currents produce magnetic fields.

Magnetism plays a fundamental role in both everyday technologies and modern physics. The relationship between electricity and magnetism was first experimentally demonstrated by Hans Christian Ørsted in the 19th century, marking the foundation of electromagnetism.

Magnetic Field

Magnetic fields are produced by moving charges and intrinsic magnetic moments of particles. A magnetic field exists around magnets, current-carrying conductors, and moving charges.

Magnetic field lines are a visualization tool: they are not physical entities. They emerge from the north pole of a magnet and terminate at the south pole, never intersecting. The density of these lines indicates field strength.

The magnetic field is a vector quantity denoted by **B**. Like poles repel, while opposite poles attract.

Magnetic Force

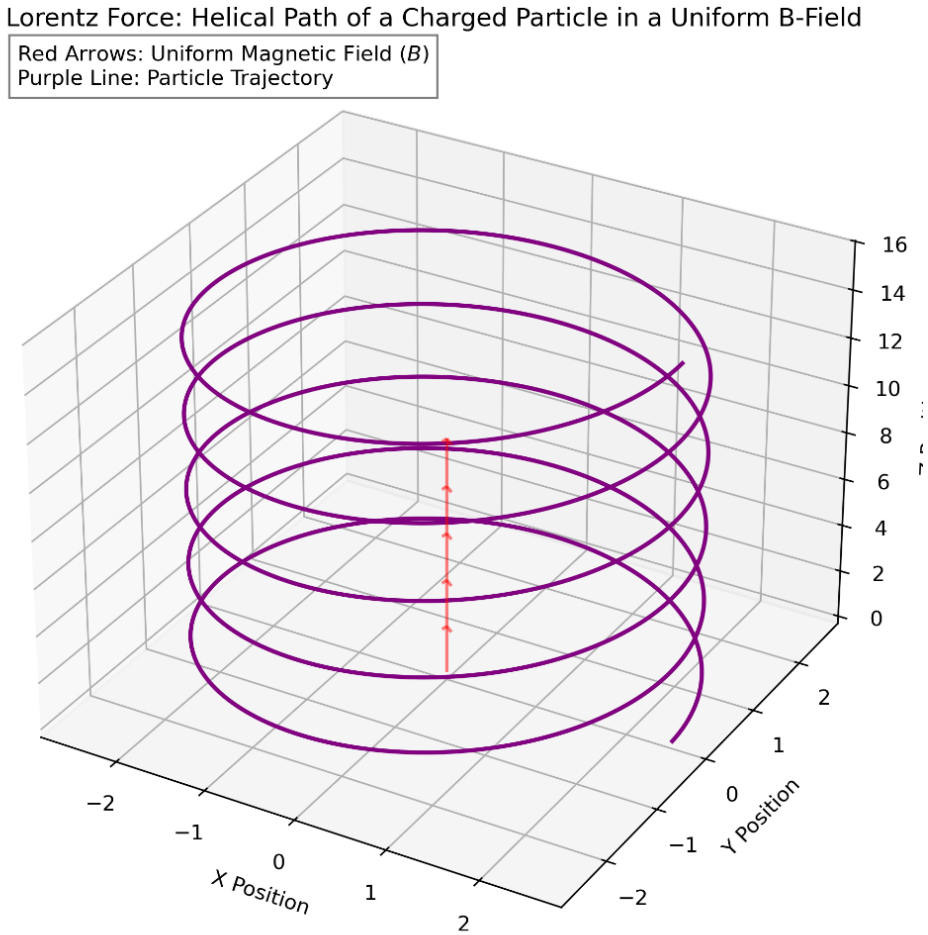
Magnetic force acts on moving charges within a magnetic field. It depends on the velocity of the particle and the magnetic field direction. The force is always perpendicular to both velocity and magnetic field, resulting in a change in direction rather than speed.

This force is fundamental in applications ranging from particle physics to electric motors.

The Lorentz force is given by:

$$F = q \cdot v \cdot B \cdot \sin(\theta)$$

Because this force is always perpendicular to velocity, charged particles follow helical trajectories in uniform magnetic fields. This behavior was simulated in the accompanying Python model, showing a particle spiraling along the field direction.



Magnetic Field Generated by Electric Current

An electric current generates a magnetic field around a conductor. Using the right-hand rule, the thumb indicates current direction, while curled fingers represent the magnetic field direction.

When a wire is coiled into a loop or solenoid, the magnetic field becomes significantly stronger and more uniform. Such structures form the basis of electromagnets and many electromechanical devices.

Faraday's Law and Lenz's Law

The relationship between electricity and magnetism is bidirectional. In 1831, Michael Faraday discovered that a changing magnetic field induces an electric current in a conductor, a phenomenon known as electromagnetic induction.

Lenz's Law states that the induced current opposes the change that produced it, reflecting the conservation of energy in electromagnetic systems.

This principle is the basis of modern generators used in hydroelectric plants, wind turbines, and nuclear power stations.

Electromagnets and Applications

Unlike permanent magnets, electromagnets generate magnetic fields only when electric current flows through them. Their field strength can be controlled by adjusting the current.

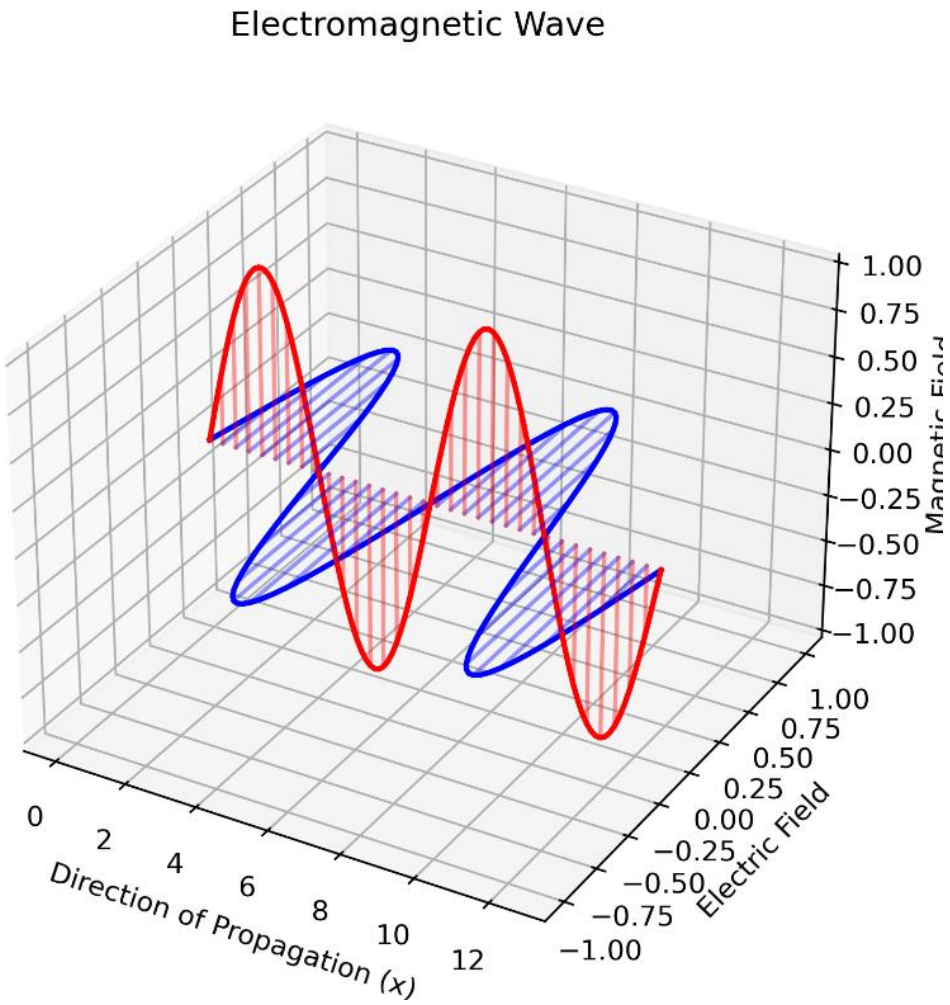
Electromagnets are fundamental to modern technology, appearing in electric motors, speakers, transportation systems, and even Earth's magnetic field interactions.

Electromagnetic Waves and Maxwell's Equations

The entire universe is permeated by electromagnetic waves, which arise from the interaction of oscillating electric and magnetic fields. The most familiar example is visible light. Other forms include radio waves, microwaves, infrared radiation, X-rays, and gamma rays.

In the 19th century, James Clerk Maxwell unified electricity and magnetism into a single theoretical framework, showing that changing electric and magnetic fields continuously generate each other, forming self-propagating waves.

These waves can travel through vacuum at the speed of light, carrying energy without requiring a material medium.



Maxwell's Equations

Maxwell unified the work of Coulomb, Faraday, and Ampère into four fundamental equations:

$\nabla \cdot E = \frac{\rho}{\varepsilon_0}$ (Gauss's Law)
$\nabla \cdot B = 0$ (Gauss's Law for Magnetism)
$\nabla \times E = \frac{\partial B}{\partial t_0}$ (Faraday's Law)
$\nabla \times B = \mu_0 J + \mu_0 \varepsilon_0 \frac{\partial E}{\partial t_0}$ (Ampe`re–Maxwell Law

Together, these equations fully describe classical electromagnetism and lead directly to the derivation of the speed of light.

Electromagnetic Spectrum

Electromagnetic waves span a broad spectrum:

- Radio waves
- Microwaves
- Infrared radiation
- Visible light
- Ultraviolet radiation

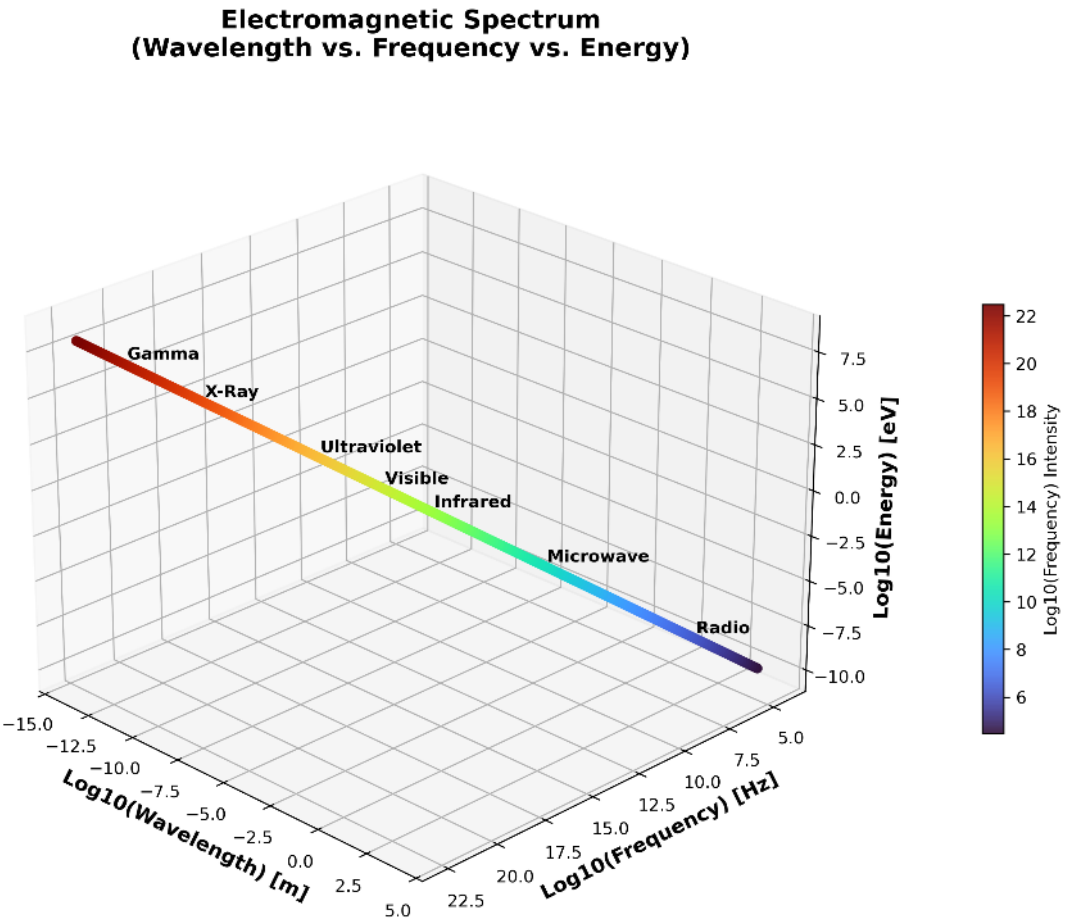
- X-rays
- Gamma rays

This spectrum underpins technologies from wireless communication to medical imaging.

Energy Distribution and Ionizing Radiation

The energy of electromagnetic waves is directly proportional to their frequency. As frequency increases (and wavelength decreases), energy increases.

Low-frequency radiation (radio and microwaves) is non-ionizing and cannot remove electrons from atoms. In contrast, high-frequency radiation such as ultraviolet, X-rays, and gamma rays is ionizing and capable of altering atomic structure.



The accompanying 3D model illustrates how decreasing wavelength leads to higher frequency and exponentially increasing photon energy. High-energy regions correspond to ionizing radiation zones.

Quantum Nature of Electromagnetism: Wave or Particle?

Classical physics describes electromagnetic waves as continuous energy fields. However, experimental evidence in the early 20th century showed that light sometimes behaves as discrete packets of energy.

Photons: Quantization of Energy

Max Planck proposed that energy is emitted in discrete quanta. Einstein later extended this idea, demonstrating that light consists of massless particles called photons.

The energy of a photon is given by:

$$E = h \cdot f$$

As frequency increases, photon energy increases accordingly.

Photoelectric Effect

The photoelectric effect provides strong evidence of the particle nature of light. When light of sufficient frequency strikes a metal surface, it ejects electrons.

Classical wave theory cannot explain why increasing intensity alone is insufficient; only frequency above a threshold can cause electron emission. This confirms that energy is transferred in discrete photon packets.

Wave–Particle Duality

Electromagnetic radiation exhibits dual behavior. It behaves as a wave in interference and diffraction phenomena, but as particles during energy exchange with matter.

This duality is a fundamental feature of quantum physics.

Quantum Electrodynamics (QED)

At a deeper level, electromagnetic interactions are described as exchanges of virtual photons between charged particles. Developed by Richard Feynman, Quantum Electrodynamics is one of the most precisely tested theories in physics.

Classical electromagnetism assumes continuous energy transfer, while experiments demonstrate quantization and discrete interactions. This discrepancy led to the development of quantum theory.

Conclusion

Electromagnetism forms the foundation of modern physics and technology, unifying electricity, magnetism, and light under a single theoretical framework. From Coulomb's Law to Maxwell's Equations and finally quantum electrodynamics, each layer of theory reveals a deeper structure of nature.

In the next part of this series, we will explore Quantum Mechanics, where probability replaces determinism and physical reality becomes fundamentally statistical.